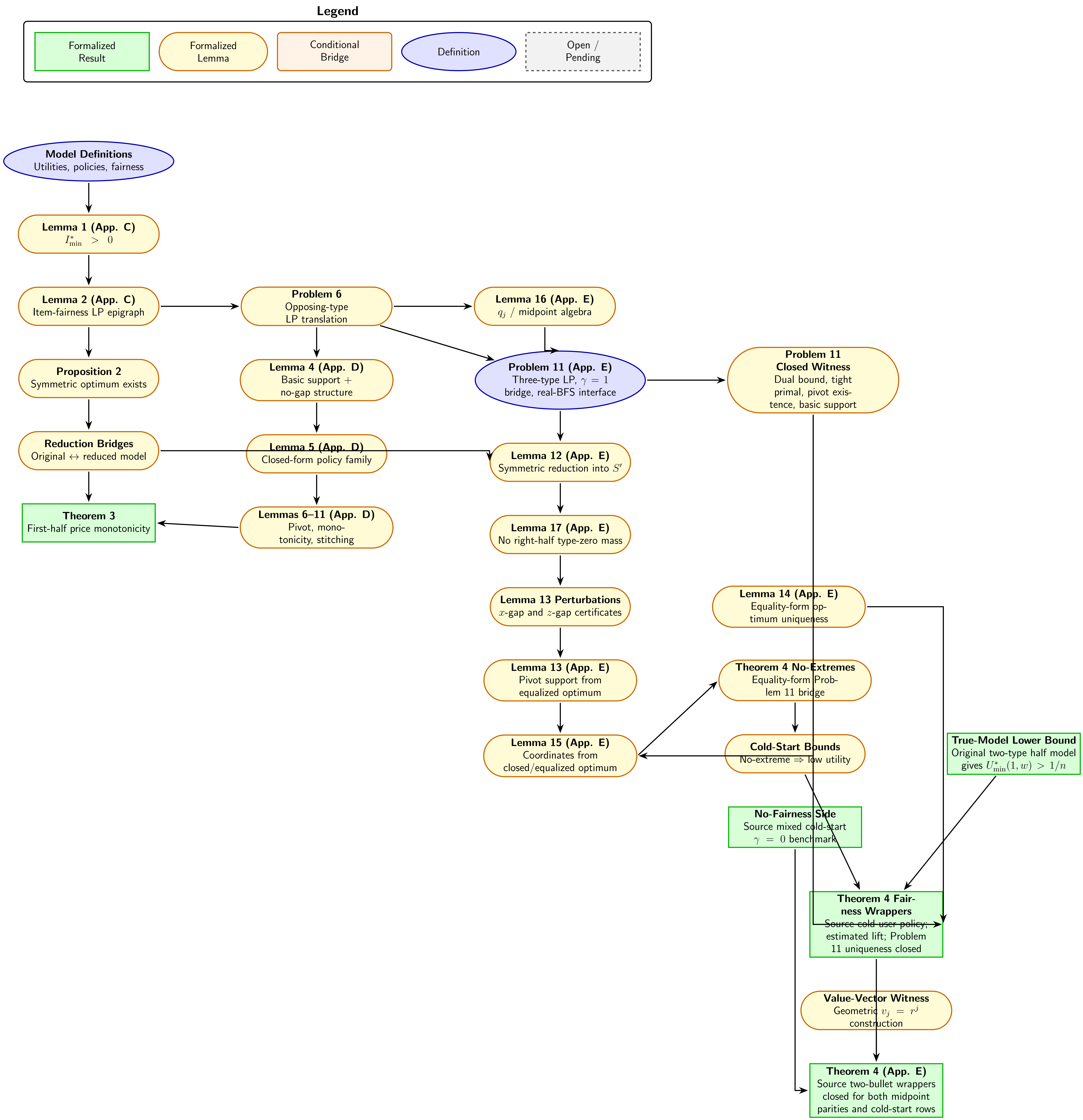
Legend
Formalized Result
Formalized Lemma
Conditional Bridge
Definition
Open / Pending
Model Definitions
Utilities, policies, fairness
Lemma 1 (App. C)
$I^*_{\min} > 0$
Lemma 2 (App. C)
Item-fairness LP epigraph
Problem 6
Opposing-type LP translation
Lemma 16 (App. E)
q_j / midpoint algebra
Proposition 2
Symmetric optimum exists
Lemma 4 (App. D)
Basic support + no-gap structure
Problem 11 (App. E)
Three-type LP, $\gamma = 1$ bridge, real-BFS interface
Problem 11 Closed Witness
Dual bound, tight primal, pivot existence, basic support
Reduction Bridges
Original $\leftrightarrow$ reduced model
Lemma 5 (App. D)
Closed-form policy family
Lemma 12 (App. E)
Symmetric reduction into S'
Theorem 3
First-half price monotonicity
Lemmas 6–11 (App. D)
Pivot, monotonicity, stitching
Lemma 17 (App. E)
No right-half type-zero mass
Lemma 13 Perturbations
x-gap and z-gap certificates
Lemma 14 (App. E)
Equality-form optimum uniqueness
Lemma 13 (App. E)
Pivot support from equalized optimum
Theorem 4 No-Extremes
Equality-form Problem 11 bridge
Lemma 15 (App. E)
Coordinates from closed/equalized optimum
Cold-Start Bounds
No-extreme $\Rightarrow$ low utility
True-Model Lower Bound
Original two-type half model gives $U^*_{\min}(1, w) > 1/n$
No-Fairness Side
Source mixed cold-start $\gamma = 0$ benchmark
Theorem 4 Fairness Wrappers
Source cold user policy; estimated lift; Problem 11 uniqueness closed
Value-Vector Witness
Geometric $v_j = r^j$ construction
Theorem 4 (App. E)
Source two-bullet wrappers closed for both midpoint parities and cold-start rows